DAILY PRACTICE CALENDAR

Grade 9 Math: Algebra 1 • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This calendar follows the **California Common Core standards for Algebra 1**, the usual Grade 9 course. Students who want an extra challenge can add **Algebra 2 Competition Math**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Expressions & Properties
2	Aug 23–Aug 27	Solving Linear Equations
3	Aug 30–Sep 3	Literal & Absolute Value Equations
4	Sep 6–Sep 10	Inequalities & Compound Inequalities
5	Sep 13–Sep 17	Functions & Function Notation
6	Sep 20–Sep 24	Domain, Range & Key Features
7	Sep 27–Oct 1	Slope & Rate of Change
8	Oct 4–Oct 8	Writing Linear Equations
9	Oct 11–Oct 15	Arithmetic Sequences
10	Oct 18–Oct 22	Review & Checkpoint 1
11	Oct 25–Oct 29	Systems: Graphing & Substitution
12	Nov 1–Nov 5	Systems: Elimination
13	Nov 8–Nov 12	Systems Word Problems
14	Nov 15–Nov 19	Inequalities in Two Variables
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Absolute Value & Piecewise Functions
16	Dec 6–Dec 10	Exponent Rules
17	Dec 13–Dec 17	Radicals & Rational Exponents
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Exponential Functions
19	Jan 10–Jan 14	Growth, Decay & Geometric Sequences
20	Jan 17–Jan 21	Mid-Year Review & Checkpoint
21	Jan 24–Jan 28	Polynomial Operations
22	Jan 31–Feb 4	Special Products
23	Feb 7–Feb 11	Factoring: GCF & Trinomials
24	Feb 14–Feb 18	Factoring $ax^2 + bx + c$ & Special Patterns
25	Feb 21–Feb 25	Solving Quadratics by Factoring
26	Feb 28–Mar 3	Quadratic Functions & Their Graphs
27	Mar 6–Mar 10	Completing the Square & Vertex Form
28	Mar 13–Mar 17	The Quadratic Formula
29	Mar 20–Mar 24	Quadratic Models & Comparing Functions
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Statistics: Center & Spread
31	Apr 10–Apr 14	Scatter Plots, Correlation & Residuals
32	Apr 17–Apr 21	Year-End Review: Equations & Functions
33	Apr 24–Apr 28	Year-End Review & Practice Test
34	May 1–May 5	Final Checkpoint & Celebration

Units 1 & 6 | Unit 2 | Unit 3 | Unit 4 | Unit 5 | Wrap-Up | Break

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WEEK 1 • Aug 16–Aug 20, 2027

Unit 1: Expressions, Equations & Inequalities

Expressions & Properties

Core: Use the properties of numbers to rewrite expressions.

Mon Aug 16

Evaluating expressions

TRY Evaluate $2x^2 - 3x + 4$ when $x = -2$.

Tue Aug 17

Distributive property

TRY Simplify $4(3a - 2) - 5a$.

Wed Aug 18

Combining like terms

TRY Simplify $3x^2 + 5x - x^2 - 8x + 2$.

Thu Aug 19

Writing expressions

TRY A phone costs $200 plus $45 a month. Write the total cost after m months.

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027

Unit 1: Expressions, Equations & Inequalities

Solving Linear Equations

Core: Simplify each side, collect the variable on one side, then isolate it.

Mon Aug 23

Multi-step equations

TRY Solve $5(x - 2) + 3 = 18$.

Tue Aug 24

Variables on both sides

TRY Solve $6x - 7 = 2x + 9$.

Wed Aug 25

Clearing fractions

TRY Solve $\frac{3x}{4} - 2 = \frac{x}{2} + 1$.

Thu Aug 26

Special cases

TRY How many solutions does $4(x - 1) = 4x - 4$ have?

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027

Unit 1: Expressions, Equations & Inequalities

Literal & Absolute Value Equations

Core: Solve a formula for any letter; an absolute value equation splits into two cases.

Mon Aug 30

Literal equations

TRY Solve $3x + 2y = 12$ for y.

Tue Aug 31

Formulas

TRY Solve $A = P(1 + rt)$ for t.

Wed Sep 1

Absolute value equations

TRY Solve $|x - 4| = 7$.

Thu Sep 2

No solution

TRY Solve $|2x + 1| = -5$.

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027

Unit 1: Expressions, Equations & Inequalities

Inequalities & Compound Inequalities

Core: Flip the sign when you multiply or divide by a negative; "and" means both, "or" means either.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Multi-step inequalities

TRY Solve $3 - 2x > 11$.

Tue Sep 7

Compound: "and"

TRY Solve $-1 < 2x + 3 \leq 9$.

Wed Sep 8

Compound: "or"

TRY Solve $x - 4 < -6$ or $3x > 12$.

Thu Sep 9

Absolute value inequalities

TRY Solve $|x - 2| \leq 5$.

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027

Unit 2: Functions & Linear Functions

Functions & Function Notation

Core: $f(x)$ is the output when the input is x.

Mon Sep 13

Function or not?

TRY Is $\{(2,5),(3,5),(4,6)\}$ a function?

Tue Sep 14

Evaluating

TRY If $f(x) = 3x - 5$, find $f(-2)$ and $f(a+1)$.

Wed Sep 15

Solving $f(x) = k$

TRY If $f(x) = 2x + 7$, for what x is $f(x) = 19$?

Thu Sep 16

Reading notation

TRY What does $f(4) = 10$ tell you about the graph of f?

Fri Sep 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027

Unit 2: Functions & Linear Functions

Domain, Range & Key Features

Core: The domain is the allowed inputs; the range is the outputs you get.

Mon Sep 20

Domain & range

TRY Find the domain and range of $\{(-1,3),(0,2),(2,3)\}$.

Tue Sep 21

Domain from a rule

TRY Find the domain of $f(x) = \sqrt{x-3}$.

Wed Sep 22

Intercepts

TRY Find the intercepts of $f(x) = 2x - 8$.

Thu Sep 23

Increasing or decreasing?

TRY Is $f(x) = -3x + 1$ increasing or decreasing?

Fri Sep 24

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027

Unit 2: Functions & Linear Functions

Slope & Rate of Change

Core: Slope = change in y ÷ change in x.

Mon Sep 27

Slope from two points

TRY Find the slope through $(-1,4)$ and $(3,-4)$.

Tue Sep 28

Rates from data

TRY A savings account has \$34 on day 2 and \$58 on day 5. What is the rate of change?

Wed Sep 29

Horizontal & vertical lines

TRY Find the slopes of $y = 5$ and $x = -2$.

Thu Sep 30

Perpendicular slopes

TRY Find the slope of a line perpendicular to $y = 4x - 1$.

Fri Oct 1

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027

Unit 2: Functions & Linear Functions

Writing Linear Equations

Core: Slope-intercept, point-slope and standard form describe the same line.

Mon Oct 4

Point-slope form

TRY Write the line with slope -3 through $(2,5)$ in slope-intercept form.

Tue Oct 5

Two points

TRY Write the line through $(-2,1)$ and $(4,13)$.

Wed Oct 6

Standard form

TRY Write $y = \frac{2}{3}x - 4$ as $Ax + By = C$ with integers.

Thu Oct 7

Parallel lines

TRY Write the line parallel to $y = -x + 3$ through $(4,1)$.

Fri Oct 8

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Functions & Linear Functions

Arithmetic Sequences **Core:** An arithmetic sequence is a linear function whose inputs are 1, 2, 3, . . .

Mon Oct 11
Common difference
TRY For 5, 12, 19, 26, . . . , find the common difference and the 10th term.

Tue Oct 12
Explicit formula
TRY Write a_n for 5, 12, 19, . . .

Wed Oct 13
Recursive formula
TRY Does $a_1 = 5$, $a_n = a_{n-1} + 7$ describe the same sequence?

Thu Oct 14
Which term?
TRY Which term of $a_n = 7n - 2$ equals 138?

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Functions & Linear Functions

Review & Checkpoint 1 **Core:** Pull Units 1–2 together and fix any weak spots.

Mon Oct 18
Review: equations
TRY Solve $2(3x - 1) = 4x + 10$.

Tue Oct 19
Review: inequalities
TRY Solve $-4x + 3 \geq 15$.

Wed Oct 20
Review: functions
TRY If $f(x) = x^2 - 2x$, find $f(-3)$.

Thu Oct 21
Review: lines
TRY Write the line through $(0, -2)$ and $(3, 4)$.

Fri Oct 22
Checkpoint 1
Units 1–2 test. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Systems & Piecewise Functions

Systems: Graphing & Substitution **Core:** The solution of a system is the point on both lines.

Mon Oct 25
Checking a solution
TRY Is $(3, -1)$ a solution of $x + y = 2$ and $2x - y = 7$?

Tue Oct 26
Substitution
TRY Solve $y = 2x - 3$ and $3x + y = 12$.

Wed Oct 27
Graphing
TRY Where do $y = x + 1$ and $y = -2x + 7$ meet?

Thu Oct 28
Number of solutions
TRY How many solutions do $y = 3x + 2$ and $6x - 2y = -4$ have?

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Systems & Piecewise Functions

Systems: Elimination **Core:** Add or subtract the equations to cancel one variable.

Mon Nov 1
Adding equations
TRY Solve $2x + y = 9$ and $3x - y = 11$.

Tue Nov 2
Multiply first
TRY Solve $x + 2y = 7$ and $3x - y = 7$.

Wed Nov 3
Multiply both
TRY Solve $2x + 3y = 8$ and $3x + 2y = 7$.

Thu Nov 4
Choosing a method
TRY Which method is quickest for $y = 4x$ and $3x + 2y = 22$? Solve it.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Systems & Piecewise Functions

Systems Word Problems — **Core:** Two unknowns need two equations.

Mon Nov 8
Mixtures
TRY Nuts at \$6/lb and raisins at \$3/lb make 12 lb of trail mix worth \$54. How many pounds of nuts?

Tue Nov 9
Break-even
TRY A company's cost is $500 + 4x$ and its revenue is $9x$. How many items must it sell to break even?

Wed Nov 10
Wind speed
TRY A plane flies 600 mi with the wind in 2 h and against the wind in 3 h. Find the plane's speed and the wind speed.

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Interest
TRY \$5,000 is split between accounts paying 3% and 5% a year. Together they earn \$190 in a year. How much is at 5%?

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Systems & Piecewise Functions

Inequalities in Two Variables — **Core:** Graph the boundary line, then shade the side that works.

Mon Nov 15
Testing points
TRY Is $(2, 1)$ a solution of $y < 3x - 4$?

Tue Nov 16
Boundary lines
TRY Is the boundary of $y \geq -x + 2$ solid or dashed?

Wed Nov 17
Systems of inequalities
TRY Is $(1, 4)$ a solution of both $y > x$ and $y \leq 5$?

Thu Nov 18
Word problems
TRY Pens cost \$2 and notebooks \$5, and you can spend at most \$30. Write an inequality.

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Systems & Piecewise Functions

Absolute Value & Piecewise Functions — **Core:** Each piece has its own rule on its own part of the domain.

Mon Nov 29
Piecewise functions
TRY $f(x) = 2x + 1$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.

Tue Nov 30
Absolute value graphs
TRY Find the vertex of $y = |x - 3| + 2$.

Wed Dec 1
Transformations
TRY How is $y = |x + 1| - 4$ moved from $y = |x|$?

Thu Dec 2
Step functions
TRY Parking costs \$5 for each hour or part of an hour. What does 2.5 hours cost?

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Exponents & Exponential Functions

Exponent Rules — **Core:** Same base: add exponents to multiply, subtract to divide, multiply for a power of a power.

Mon Dec 6
Product rule
TRY Simplify $x^3 \cdot x^5$.

Tue Dec 7
Power rules
TRY Simplify $(2a^2)^3$.

Wed Dec 8
Negative exponents
TRY Simplify $\dfrac{6x^{-2}y^3}{3x \cdot y^{-1}}$ using positive exponents.

Thu Dec 9
Zero exponents
TRY Compute $(5x^4y)^0 + 3^{-1}$.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Exponents & Exponential Functions

Radicals & Rational Exponents

Core: $a^{1/n}$ is the nth root of a.

Mon Dec 13
Simplifying radicals
TRY Simplify $\sqrt{72}$.

Tue Dec 14
Rational exponents
TRY Compute $27^{2/3}$.

Wed Dec 15
Adding radicals
TRY Simplify $\sqrt{12} + \sqrt{75}$.

Thu Dec 16
Rationalizing
TRY Simplify $\dfrac{10}{\sqrt{5}}$.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Exponents & Exponential Functions

Exponential Functions

Core: In $f(x) = a \cdot b^x$, a is the starting value and b is the growth factor.

Mon Jan 3
Evaluating
TRY If $f(x) = 3 \cdot 2^x$, find $f(4)$.

Tue Jan 4
Growth or decay?
TRY Does $y = 500(0.8)^x$ show growth or decay? By what percent per step?

Wed Jan 5
From a table
TRY A table has x: 0, 1, 2 and y: 4, 12, 36. Write the function.

Thu Jan 6
Linear vs. exponential
TRY Which eventually becomes larger: $100x$ or 2^x?

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Exponents & Exponential Functions

Growth, Decay & Geometric Sequences

Core: A constant percent change means an exponential model.

Mon Jan 10
Compound interest
TRY $2,000 earns 5% a year, compounded yearly. How much is there after 3 years?

Tue Jan 11
Depreciation
TRY A $20,000 car loses 15% of its value each year. What is it worth after 2 years?

Wed Jan 12
Geometric sequences
TRY Find the 6th term of 3, 12, 48, . . .

Thu Jan 13
Half-life
TRY 80 g of a substance has a half-life of 5 days. How much is left after 15 days?

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Exponents & Exponential Functions

Mid-Year Review & Checkpoint

Core: Review Units 1–4 and practice working steadily through a full test.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: systems
TRY Solve $x + y = 12$ and $x - y = 2$.

Tue Jan 18
Review: exponent rules
TRY Simplify $(3x^2)^2 \cdot x$.

Wed Jan 19
Review: exponential models
TRY By what percent does $y = 200(1.1)^x$ grow each step?

Thu Jan 20
Review: absolute value
TRY Solve $|x - 1| = 4$.

Fri Jan 21
Mid-Year Checkpoint
Units 1–4 test. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Polynomials, Factoring & Parabolas

Polynomial Operations — **Core:** Combine like terms; to multiply, multiply every term by every term.

Mon Jan 24
Adding & subtracting
TRY Simplify $(3x^2 - 2x + 5) - (x^2 + 4x - 1)$.

Tue Jan 25
Monomial × polynomial
TRY Expand $2x(3x^2 - x + 4)$.

Wed Jan 26
Binomial × binomial
TRY Expand $(x + 4)(2x - 3)$.

Thu Jan 27
Degree
TRY What is the degree of $5x^3 - x^4 + 2$?

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Polynomials, Factoring & Parabolas

Special Products — **Core:** $(a + b)^2 = a^2 + 2ab + b^2$ and $(a + b)(a - b) = a^2 - b^2$.

Mon Jan 31
Square of a sum
TRY Expand $(x + 5)^2$.

Tue Feb 1
Square of a difference
TRY Expand $(3x - 2)^2$.

Wed Feb 2
Difference of squares
TRY Expand $(2x + 7)(2x - 7)$.

Thu Feb 3
Mental math
TRY Use a special product to compute 41×39.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Polynomials, Factoring & Parabolas

Factoring: GCF & Trinomials — **Core:** Always take out the greatest common factor first.

Mon Feb 7
Greatest common factor
TRY Factor $12x^3 - 18x^2$.

Tue Feb 8
$x^2 + bx + c$
TRY Factor $x^2 + 7x + 12$.

Wed Feb 9
Negative constant
TRY Factor $x^2 - 2x - 15$.

Thu Feb 10
Grouping
TRY Factor $x^3 + 3x^2 + 2x + 6$.

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Polynomials, Factoring & Parabolas

Factoring $ax^2 + bx + c$ & Special Patterns — **Core:** Use the ac method; look for perfect squares and differences of squares.

Mon Feb 14
The ac method
TRY Factor $3x^2 + 10x + 8$.

Tue Feb 15
Difference of squares
TRY Factor $x^2 - 81$.

Wed Feb 16
Perfect-square trinomials
TRY Factor $4x^2 + 12x + 9$.

Thu Feb 17
Factor completely
TRY Factor $2x^3 - 8x$ completely.

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Polynomials, Factoring & Parabolas

Solving Quadratics by Factoring

Core: If a product is 0, one of the factors must be 0.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Zero-product property

TRY Solve $(x-4)(2x+1)=0$.

Tue Feb 22

Factor, then solve

TRY Solve $x^2 - x - 12 = 0$.

Wed Feb 23

Rearrange first

TRY Solve $x^2 = 6x$.

Thu Feb 24

Word problems

TRY A rectangle has area 48, and its length is 2 more than its width. Find its dimensions.

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 5: Polynomials, Factoring & Parabolas

Quadratic Functions & Their Graphs

Core: A parabola has a vertex, an axis of symmetry and intercepts.

Mon Feb 28

Vertex & axis

TRY Find the vertex and axis of symmetry of $y = x^2 - 6x + 5$.

Tue Feb 29

Intercepts

TRY Find the zeros and the y-intercept of $y = x^2 - 6x + 5$.

Wed Mar 1

Up or down?

TRY Does $y = -2x^2 + 3$ open up or down? What is its maximum or minimum?

Thu Mar 2

Projectiles

TRY A ball's height is $h(t) = -16t^2 + 64t$ feet. What is its greatest height?

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Quadratic Equations & Statistics

Completing the Square & Vertex Form

Core: Add $\left(\frac{b}{2}\right)^2$ to make a perfect square.

Mon Mar 6

Making a perfect square

TRY What number completes the square: $x^2 + 8x + \square$?

Tue Mar 7

Vertex form

TRY Write $x^2 + 8x + 10$ in vertex form and give the vertex.

Wed Mar 8

Solving

TRY Solve $x^2 + 6x - 7 = 0$ by completing the square.

Thu Mar 9

Square roots

TRY Solve $(x-3)^2 = 25$.

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Quadratic Equations & Statistics

The Quadratic Formula

Core: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; the discriminant $b^2 - 4ac$ counts the real solutions.

Mon Mar 13

Using the formula

TRY Solve $x^2 - 4x - 5 = 0$.

Tue Mar 14

Irrational solutions

TRY Solve $x^2 + 2x - 4 = 0$.

Wed Mar 15

The discriminant

TRY How many real solutions does $2x^2 - 3x + 5 = 0$ have?

Thu Mar 16

Choosing a method

TRY Which method is quickest for $x^2 - 49 = 0$? Solve it.

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Quadratic Equations & Statistics

Quadratic Models & Comparing Functions

Core: Linear: equal differences. Quadratic: equal second differences. Exponential: equal ratios.

Mon Mar 20
Second differences
TRY Which model fits y: 1, 4, 9, 16, 25?

Tue Mar 21
Equal ratios
TRY Which model fits y: 3, 6, 12, 24?

Wed Mar 22
Maximum revenue
TRY Revenue is $R = x(50 - x)$. What is the greatest possible revenue?

Thu Mar 23
Where graphs meet
TRY Solve $x^2 = 2x + 3$.

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Quadratic Equations & Statistics

Statistics: Center & Spread

Core: Use mean and standard deviation for symmetric data, median and IQR for skewed data.

Mon Apr 3
Mean & median
TRY Find the mean and median of 3, 5, 7, 7, 8, 12.

Tue Apr 4
Interquartile range
TRY Find the IQR of 2, 4, 6, 8, 10, 12, 14, 16.

Wed Apr 5
Outliers
TRY Data have Q1 $= 10$ and Q3 $= 20$. By the $1.5\times$ IQR rule, is 38 an outlier?

Thu Apr 6
Comparing spread
TRY Which set has the larger standard deviation: $\{5, 5, 5, 5\}$ or $\{1, 5, 9\}$?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Quadratic Equations & Statistics

Scatter Plots, Correlation & Residuals

Core: r measures the strength of a linear pattern; residual $=$ actual $-$ predicted.

Mon Apr 10
Correlation
TRY What does $r = -0.92$ tell you?

Tue Apr 11
Residuals
TRY A model is $y = 2x + 3$. The data point is $(4, 13)$. Find the residual.

Wed Apr 12
Interpreting slope
TRY A line of fit for price vs. weight (lb) is $y = 1.5x + 20$. What does 1.5 mean?

Thu Apr 13
Correlation vs. causation
TRY Ice cream sales and sunburns are correlated. Does one cause the other?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Year-End Review

Year-End Review: Equations & Functions

Core: Mix the skills from Units 1–4 and show your work.

Mon Apr 17
Review: inequalities
TRY Solve $|x + 1| > 3$.

Tue Apr 18
Review: systems
TRY Solve $3x + y = 10$ and $x - y = 2$.

Wed Apr 19
Review: exponential growth
TRY $1,000 earns 4% a year, compounded yearly. How much is there after 2 years?

Thu Apr 20
Timed set: 10 in 20 min
TRY Factor $x^2 - 9x + 20$.

Fri Apr 21
Weekly Quiz
10 mixed review problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Year-End Review

Year-End Review & Practice Test

Core: Practice a full test, then learn from every error.

Mon Apr 24

Review: quadratics

TRY Solve $x^2 + 4x - 21 = 0$.

Tue Apr 25

Review: radicals

TRY Simplify $\sqrt{50} - \sqrt{8}$.

Wed Apr 26

Review: statistics

TRY A data point is 20, and the model predicts 23. Find the residual.

Thu Apr 27

Practice Test

Full-length Algebra 1 practice test covering the whole year.

Fri Apr 28

Error-log review

Redo every missed problem from the test; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Year-End Review

Final Checkpoint & Celebration

Core: Show what you have learned and get ready for Geometry.

Mon May 1

Light review

TRY Find the vertex of $y = (x - 2)^2 + 5$.

Tue May 2

Mixed warm-up set

TRY If $f(x) = 2^x + 3$, find $f(3)$.

Wed May 3

Final Checkpoint

Covers the whole year.

Thu May 4

Checkpoint review

Go over solutions together; students explain one problem each.

Fri May 5

Celebration & reflection

Certificates, progress review, and goals for Geometry next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 18
Tue: $7a - 8$
Wed: $2x^2 - 3x + 2$
Thu: $200 + 45m$

Week 2
Mon: $x = 5$
Tue: $x = 4$
Wed: $x = 12$
Thu: Infinitely many

Week 3
Mon: $y = -\frac{3}{2}x + 6$
Tue: $t = \frac{A - P}{P \cdot r}$
Wed: $x = 11$ or $x = -3$
Thu: No solution

Week 4
Mon: $x < -4$
Tue: $-2 < x \leq 3$
Wed: $x < -2$ or $x > 4$
Thu: $-3 \leq x \leq 7$

Week 5
Mon: Yes
Tue: -11 and $3a - 2$
Wed: $x = 6$
Thu: The point $(4, 10)$ is on it

Week 6
Mon: Domain $\{-1, 0, 2\}$; range $\{2, 3\}$
Tue: $x \geq 3$
Wed: $(4, 0)$ and $(0, -8)$
Thu: Decreasing

Week 7
Mon: -2
Tue: $8 per day
Wed: 0; undefined
Thu: $-\frac{1}{4}$

Week 8
Mon: $y = -3x + 11$
Tue: $y = 2x + 5$
Wed: $2x - 3y = 12$
Thu: $y = -x + 5$

Week 9
Mon: 7; 68
Tue: $a_n = 7n - 2$
Wed: Yes
Thu: The 20th

Week 10
Mon: $x = 6$
Tue: $x \leq -3$
Wed: 15
Thu: $y = 2x - 2$

Week 11
Mon: Yes
Tue: $(3, 3)$
Wed: $(2, 3)$
Thu: Infinitely many (same line)

Week 12
Mon: $(4, 1)$
Tue: $(3, 2)$
Wed: $(1, 2)$
Thu: Substitution; $(2, 8)$

Week 13
Mon: 6 lb
Tue: 100
Wed: Plane 250 mph, wind 50 mph
Thu: $2,000

Week 14
Mon: Yes $(1 < 2)$
Tue: Solid
Wed: Yes
Thu: $2p + 5n \leq 30$

Week 15
Mon: -5 and 9
Tue: $(3, 2)$
Wed: Left 1, down 4
Thu: $15

Week 16
Mon: x^8
Tue: $8a^6$
Wed: $\frac{2y^4}{x^3}$
Thu: $\frac{4}{3}$

Week 17
Mon: $6\sqrt{2}$
Tue: 9
Wed: $7\sqrt{3}$
Thu: $2\sqrt{5}$

Week 18
Mon: 48
Tue: Decay, 20% per step
Wed: $y = 4 \cdot 3^x$
Thu: 2^x

Week 19
Mon: $2,315.25
Tue: $14,450
Wed: 3,072
Thu: 10 g

Week 20
Mon: $(7, 5)$
Tue: $9x^5$
Wed: 10%
Thu: $x = 5$ or $x = -3$

Week 21
Mon: $2x^2 - 6x + 6$
Tue: $6x^3 - 2x^2 + 8x$
Wed: $2x^2 + 5x - 12$
Thu: 4

Week 22
Mon: $x^2 + 10x + 25$
Tue: $9x^2 - 12x + 4$
Wed: $4x^2 - 49$
Thu: 1,599

Week 23
Mon: $6x^2(2x - 3)$
Tue: $(x + 3)(x + 4)$
Wed: $(x - 5)(x + 3)$
Thu: $(x^2 + 2)(x + 3)$

Week 24
Mon: $(3x + 4)(x + 2)$
Tue: $(x + 9)(x - 9)$
Wed: $(2x + 3)^2$
Thu: $2x(x + 2)(x - 2)$

Week 25
Mon: $x = 4$ or $x = -\frac{1}{2}$
Tue: $x = 4$ or $x = -3$
Wed: $x = 0$ or $x = 6$
Thu: 6 by 8

Week 26
Mon: Vertex $(3, -4)$; $x = 3$
Tue: Zeros 1 and 5; y-intercept 5
Wed: Down; maximum 3
Thu: 64 ft

Week 27
Mon: 16
Tue: $(x + 4)^2 - 6$; $(-4, -6)$
Wed: $x = 1$ or $x = -7$
Thu: $x = 8$ or $x = -2$

Week 28
Mon: $x = 5$ or $x = -1$
Tue: $x = -1 \pm \sqrt{5}$
Wed: None
Thu: Square roots; $x = \pm 7$

Week 29
Mon: Quadratic
Tue: Exponential
Wed: 625 (at $x = 25$)
Thu: $x = 3$ or $x = -1$

Week 30
Mon: Mean 7, median 7
Tue: 8
Wed: Yes (the upper fence is 35)
Thu: $\{1, 5, 9\}$

Week 31
Mon: A strong negative linear relationship
Tue: 2
Wed: Each extra pound adds about $1.50
Thu: No: hot weather causes both

Week 32
Mon: $x < -4$ or $x > 2$
Tue: $(3, 1)$
Wed: $1,081.60
Thu: $(x - 4)(x - 5)$

Week 33
Mon: $x = 3$ or $x = -7$
Tue: $3\sqrt{2}$
Wed: -3

Week 34
Mon: $(2, 5)$
Tue: 11

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